National Taiwan Ocean University
Department of Computer Science

Computer Networks — 2025 Midterm Examination

Scope: Chapter 1 ~ Chapter 3

Instructions: *This is a closed-book exam. Answer all questions. Write clearly and justify your answers where required. Total: 100 points.*

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1. (15 points) Internet Protocol Stack

Describe the five-layer Internet protocol stack. For each layer, explain its main function and give at least one example of a protocol used at that layer.

(Hint: The five layers are — Application, Transport, Network, Link, Physical.)

2. (15 points) Packet Delay

(a) Briefly define each of the four types of packet delay that a packet experiences as it travels through a router/node:

1. Nodal Processing Delay: ____
2. Queuing Delay: ____
3. Transmission Delay: ____
4. Propagation Delay: ____

(b) Among the four types, which one is most likely to cause extremely long delays under heavy network load, and why?

(c) A packet has a size of L = 40,000 bits. It is transmitted over a link with a bandwidth of R = 3 Mbps. What is the transmission delay?

3. (25 points) Circuit Switching vs. Packet Switching

(a) Explain how **circuit switching** works. What are its main characteristics, and what are the drawbacks compared to packet switching?

(b) Explain how **packet switching** works. What are its advantages over circuit switching?

Suppose users share a **1 Mbps** link. Each user requires **100 kbps** when actively transmitting, but each user transmits only **10%** of the time.

(c) If **circuit switching** is used, how many users can be supported simultaneously?

(d) Now suppose **packet switching** is used with **40 users**. Using the binomial distribution, write an expression for the probability that exactly *n* users are transmitting simultaneously at any given time.

(e) Find the probability that there are **11 or more** users transmitting simultaneously. (You may leave your answer in summation form.)

(f) In packet switching, a router uses **store-and-forward** transmission. Explain what this means — why must the router wait until the entire packet arrives before forwarding it?

(g) What problem might arise if a router forwarded each bit immediately upon arrival, without waiting for the complete packet?

4. (10 points) DNS Resource Records

For each of the following descriptions, identify the type of DNS resource record it corresponds to. Choose from: **A, NS, MX, CNAME**.

#	Description	Record Type
(a)	Maps a mail server hostname for a domain (e.g., directs email for foo.com to mail.foo.com)	________
(b)	Provides an alias for a canonical hostname (e.g., www.foo.com → relay1.bar.foo.com)	________
(c)	Maps a hostname directly to its IPv4 address	________
(d)	Identifies the authoritative name server for a domain	________
(e)	The record returned by an authoritative DNS server that provides the IP address of a host	________
(f)	The record type stored in a root or TLD DNS server pointing to the next DNS server in the hierarchy	________

5. (10 points) Network Performance Metrics for Different Applications

For each of the following network applications, discuss which performance metrics (choose from: **throughput, delay, packet loss, security**) are most critical. Give a brief explanation for each choice.

(a) An online banking / financial transaction system:

(b) A short-video streaming platform (such as TikTok / Reels):

(c) The DNS system (especially when it becomes a target of DDoS attacks):

6. (13 points) Reliable Data Transfer (RDT 3.0)

(a) (9 pts) The figure below shows a simplified RDT 3.0 sender finite-state machine with several labels missing. Fill in the blanks (labeled 1–9) with the correct actions or conditions from the list below:

Available terms: corrupt(rcvpkt), isACK(rcvpkt,0), isACK(rcvpkt,1), notCorrupt(rcvpkt), udt_send(sndpkt), start_timer, stop_timer, timeout, sndpkt = make_pkt(0,data,checksum), sndpkt = make_pkt(1,data,checksum)

Blank [1]: Condition to stay in 'Wait for ACK 0' state when a bad/wrong ACK arrives

Answer: ____

Blank [2]: Condition: correct ACK for packet 0 received

Answer: ____

Blank [3]: Action: resend the current packet

Answer: ____

Blank [4]: Action: start the retransmission timer

Answer: ____

Blank [5]: Condition: packet received is not corrupted

Answer: ____

Blank [6]: Action: create packet 1 with data and checksum

Answer: ____

Blank [7]: Event that triggers retransmission (timer expires)

Answer: ____

Blank [8]: Condition: packet received without corruption (in 'Wait for ACK 1')

Answer: ____

Blank [9]: Condition: correct ACK for packet 1 received

Answer: ____

(b) (4 pts) Trace through the following scenario using RDT 3.0. Describe what happens at each step:

Step 1: Sender transmits packet P0. Receiver gets P0 correctly and sends ACK0.

Step 2: Sender transmits packet P1. The ACK1 from receiver is delayed (not yet received by sender).

Step 3: Sender's timer expires. Sender retransmits P1. Receiver now receives two copies of P1.

Step 4: What does the receiver do upon receiving the duplicate P1? What does it reply?

7. (7 points) Pipelined Protocols — Go-Back-N / Selective Repeat

Consider a scenario where three senders A, B, C each need to send data frames. Suppose the window size is 2 and sequence numbers cycle (0, 1). Frames from sender A are labeled A0, A1; from B are B0, B1; from C are C0, C1.

(a) Trace the correct frame-sending order in the sequence below assuming Go-Back-N is used, and some frames are lost and must be retransmitted. Fill in the blank frames in the sequence:

A0, A1, B0, A1, B___, A1, C0, C1, C0

(b) Explain why A1 appears multiple times in the sequence above. Which protocol behavior does this illustrate?

8. (5 points) TCP Basics

(a) When you open a browser and type a URL (e.g., https://tronclass.ntou.edu.tw), which application-layer protocol is used to retrieve the web page?

(b) What is the minimum size (in bytes) of a TCP segment header?

(c) In a TCP segment, the **Acknowledgement Number** field is set to 1. What does this value indicate?

(d) What does it mean when the **Receive Window** field in a TCP segment is set to 0? What should the sender do?

(e) How can a TCP receiver determine where the TCP header ends and the data payload begins in a received segment? Which field in the TCP header provides this information?